Introduction to Research project: S-matrix constraints using non-invertible symmetries

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1. Motivations: generalised symmetries and constraints on dynamics

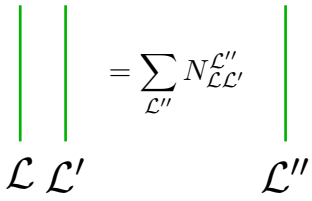
Symmetries are perhaps the most fundamental principles that guide the study of physics, from the basics of classical mechanics to the classification of particles in the standard model. The term "symmetry of the system" can be used in a more or less specific sense, but generally we say that a system has a certain symmetry if applying a specific set of transformations, called symmetry transformations, to the system, it "looks the same". Until the last century, the mathematical language of symmetry transformations was equivalent to that of groups, and their representations over a particular set of objects (that can be a vector space, a manifold or something else depending on the context). For example, in the case of Lagrangian formalism in classical field theory, the system is said to be invariant under a certain symmetry group G if the fields form a representation of the group on which they act, and if after this action the Lagrangian remains invariant. In recent decades, many examples of new kinds of transformations, that cannot be described in terms of a group action, were found. These kinds of transformations are now called *generalised symmetries*. In this review we focus on a specific class of generalised symmetries, called *non-invertible symmetries*. As the name suggests, they are characterised by an algebra in which the elements do not have an inverse, of the type:

$$\mathcal{L} \times \mathcal{L}' = \sum_{\mathcal{L}''} N_{\mathcal{L}\ \mathcal{L}'}^{\mathcal{L}''} \mathcal{L}''. \tag{1.1}$$

Here, $\mathcal{L}, \mathcal{L}', \mathcal{L}''$ are elements of the algebra and $N_{\mathcal{L}\ \mathcal{L}'}^{\mathcal{L}''}$ are non-negative integers. Non-invertible symmetries find applications in various contexts, from basic models in statistical physics to the standard model in particle physics [1].

In addition to serving as a means of characterising systems, symmetries also constitute a tool for obtaining information about them. In particular they play a major role in the bootstrap approaches, where one tries to infer information of systems neglecting the details of a particular model and focusing on the consequences of general principles, such as causality, unitarity, particle content and symmetries. There are different kinds of bootstrap approaches in QFT. Here we focus on the S-matrix bootstrap, where by using fundamental principles one can try to constrain the S-matrix. The S-matrix bootstrap has a long history, but only recently [2] [3] non-invertible symmetries were introduced to this approach. These notes are a review of these articles, in which we try to make explicit the bridge between non-invertible symmetries and their realization in different types of S-matrix bootstrap approaches. The review is structured as follows:

- In Section 2, we introduce the basics of non-invertible symmetries in a 1+1D QFT, and their action on the Hilbert space generated by kinks.
- In Section 3, we analyse the consequences of these symmetries on the dynamics, in particular deriving the Ward identities and modified crossing relation, in the context of integrable models.
- Finally, in Section 4, we use the SymTFT formalism in order to implement the non-invertible symmetry in numerical S-matrix bootstrap, giving explicit examples.



(2.1)

Figure 1: Fusion rule of symmetry lines.

2. Non-invertible symmetries

2.1 Fusion rules and F-symbols

As mentioned in the last section, we start by introducing topological operators in a 1+1D QFT, from an operational point of view. A brief mathematical introduction can be found in Appendix A, whereas for details see [4]. In a 1+1D QFT, a topological operator $\mathcal{L}$ is a one (spacetime) dimensional operator whose action does not depend on its precise geometry, but only on its topology. Such an operator is called a *symmetry* if it commutes with the stress-energy tensor $T_{\mu\nu}$ of the theory. Now let's consider a theory with a finite number of these topological lines. We require that they satisfy a *fusion rule*, of the type (2.2):

$$\mathcal{L} \times \mathcal{L}' = \sum_{\mathcal{L}''} N_{\mathcal{L}\ \mathcal{L}'}^{\mathcal{L}''} \mathcal{L}''. \tag{2.2}$$

Here, $N_{\mathcal{L}\ \mathcal{L}'}^{\mathcal{L}''} \in \mathbb{N}$ are non-negative integers. For simplicity we will assume $N_{\mathcal{L}\ \mathcal{L}'}^{\mathcal{L}''} = 0, 1$. We can now make three observations.

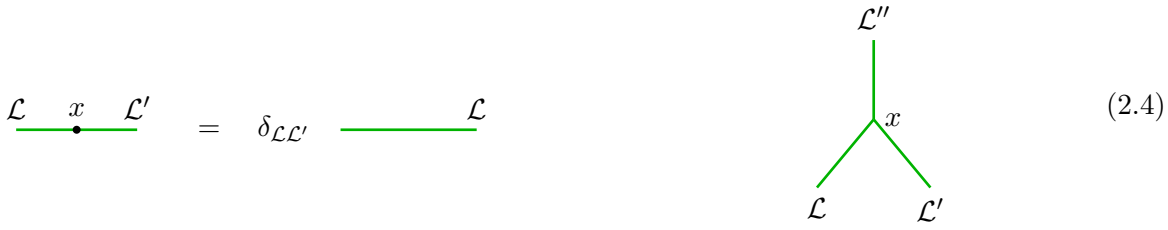
1. This type of fusion rule is similar to OPE between local operators, often used in CFTs. However, in our case there is no position dependence, nor problems related to the radius of convergence. Also in this sense these line operators are topological.

2. The topological lines do not satisfy a group-like relation, but they (almost) form a ring with the $(+, \times)$ operations. We always have the identity line $\mathbb{I}$, while an inverse line usually does not exist. We call a line $\mathcal{L}'$ inverse of $\mathcal{L}$ if $N_{\mathcal{L}\ \mathcal{L}'}^{\mathbb{I}} = 1$. A line $\mathcal{L}$ is said to be *self-dual* if $N_{\mathcal{L}\ \mathcal{L}}^{\mathbb{I}} = 1$. For simplicity, we will consider only self-dual lines.

3. The $+$ operation should be understood as an operation on correlation functions.

$$\langle (\mathcal{L} + \mathcal{L}')... \rangle = \langle \mathcal{L}... \rangle + \langle \mathcal{L}'... \rangle \,. \tag{2.3}$$

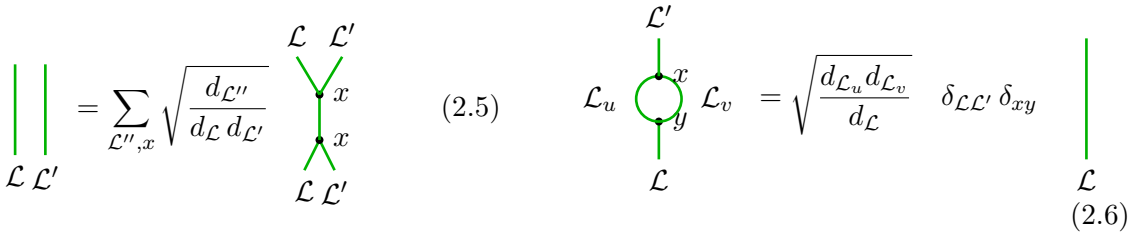
Two, three or more lines can be attached to one local operator, forming junctions. These operators form a vector space, which can be represented as a linear combination of a basis. In particular, we can choose a basis in which we use only *simple lines*, whose double junctions satisfy the first relation in Figure 2.



(2.4)

Figure 2: Double and triple junctions.

Considering the trivalent topological junctions, it can be proved that the dimension of the vector space of topological junctions is $\dim(V_{\mathcal{L}\ \mathcal{L}'}^{\mathcal{L}''}) = N_{\mathcal{L}\ \mathcal{L}'}^{\mathcal{L}''}$. By choosing a specific basis of the junctions, we have the orthogonality and completeness relations shown in Figure 3.

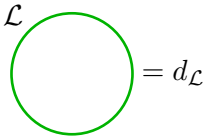


(2.5)

(2.6)

Figure 3: Completeness and orthogonality conditions of lines.

Having fixed the conventions, one can start with computing quantities of the theory. The first one is called the *quantum dimension* $d_{\mathcal{L}}$, defined as the expectation value of a loop of the line $\mathcal{L}$. In a unitary theory one can prove that $d_{\mathcal{L}} \geq 1$ [4]. Furthermore, the relation (2.8) holds.

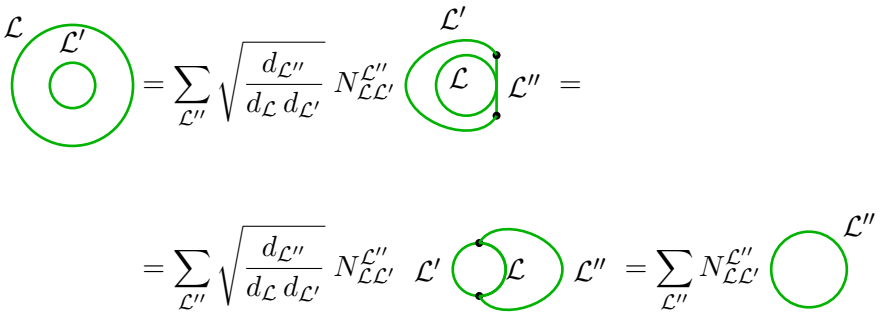


(2.7)

Figure 4: Diagrammatic representation of the quantum dimension $d_{\mathcal{L}}$.

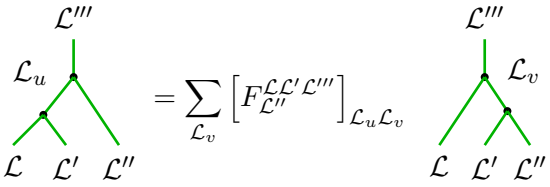
$$d_{\mathcal{L}} d_{\mathcal{L}'} = \sum_{\mathcal{L}''} N_{\mathcal{L}\ \mathcal{L}'}^{\mathcal{L}''} d_{\mathcal{L}''}. \tag{2.8}$$

A diagrammatic proof is shown in Figure 5. Other than the fusion rules, in order to fix the non-invertible symmetry one has to define the *F-symbols*. Using the associativity of the fusion product $(\mathcal{L} \times \mathcal{L}') \times \mathcal{L}'' = \mathcal{L} \times (\mathcal{L}' \times \mathcal{L}'')$, the F-symbols are defined diagrammatically as:



(2.9)

Figure 5: Pictorial proof of the relation $d_{\mathcal{L}} d_{\mathcal{L}'} = \sum_{\mathcal{L}''} N_{\mathcal{L}\ \mathcal{L}'}^{\mathcal{L}''} d_{\mathcal{L}''}$.



(2.10)

Figure 6: Definition of F-symbols.

Such symbols are not completely arbitrary (even if they present some gauge freedom), but they are constrained by the *pentagon identity*, coming from two different ways of calculating the same diagram. The F-symbols together with fusion rules define the *fusion category* $\mathcal{C}$.

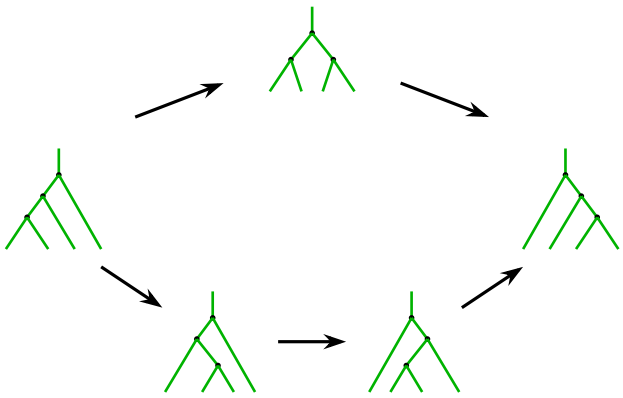


Figure 7: Pentagon identity of F-symbols.

2.2 An explicit example: Fibonacci category

In order to see the last definitions in action, we perform some explicit calculations with the simplest non-trivial example of fusion category: the *Fibonacci category*, $\mathcal{C} = Fib$. This category contains two objects $\mathbb{I}$ and τ, with the fusion rules:

$$\mathbb{I} \times \mathbb{I} = \mathbb{I}, \quad \tau \times \mathbb{I} = \mathbb{I} \times \tau = \tau, \quad \tau \times \tau = \mathbb{I} + \tau. \tag{2.11}$$

The last equation implies that this category does not form a group. By this relation we instantly get $d_{\mathbb{I}} = 1$, $d_\tau^2 = d_\tau + 1$. We start by fixing some gauge, requiring that all the F-symbols with at least an $\mathbb{I}$ as input or as output are equal to one. This makes sense thinking about the fact that $\mathbb{I}$ is indeed the identity line.

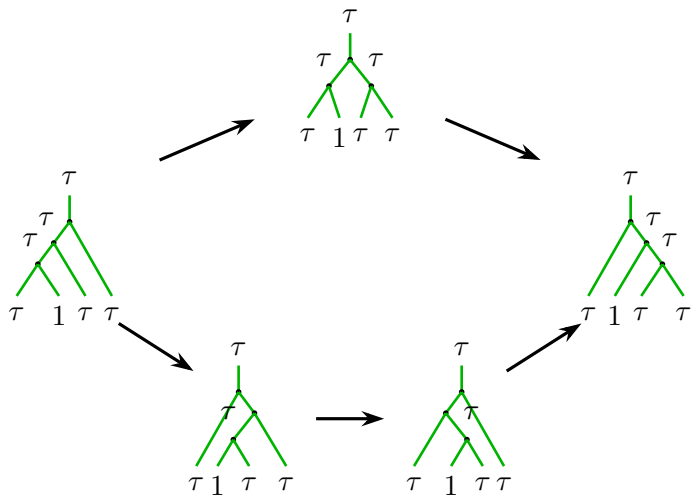


Figure 8: Example of pentagon equation containing one $\mathbb{I}$.

So it remains to calculate $[F_\tau^{\tau\tau\tau}]_{**}$, with $* = \mathbb{I}, \tau$. The steps to use the pentagon equations are:

- Choose a middle line in the starting and the final diagrams;
- Derive the intermediate diagrams compatible with the previous choice;
- Apply the pentagon equation projected to the chosen final diagram.

Following the procedure, one deduces a series of polynomial equations in the F-symbols, some of which are trivial and others redundant. We define

$$A = [F_\tau^{\tau\tau\tau}]_{11}, \quad B = [F_\tau^{\tau\tau\tau}]_{1\tau}, \quad C = [F_\tau^{\tau\tau\tau}]_{\tau 1}, \quad D = [F_\tau^{\tau\tau\tau}]_{\tau\tau}.$$

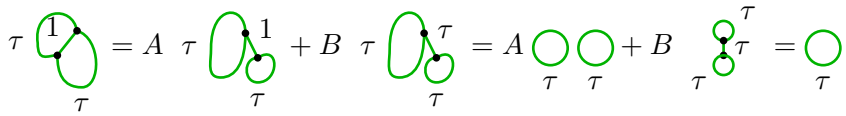
We have the following independent equations:

$$A = BC, \quad 1 = A + D^2, \quad 0 = A^2 + BCD, \quad D^2 = BC + D^3. \tag{2.12}$$

As solutions we obtain:

$$A = BC = -D, \quad D^2 = D + 1. \tag{2.13}$$

The solution related to the positive quantum dimension can be obtained from the relation $A = 1/d_\tau$, proved diagrammatically



(2.14)

Figure 9: Proof of $A = 1/d_\tau$.

The diagram following the coefficient B vanishes by the vanishing tadpole theorem [4]. We note that, even after this observation, we have some gauge freedom. In fact, B and C appear only through their product BC, and it is not possible to determine them independently. A common choice is the symmetric gauge, $B = C = 1/\sqrt{\phi}$. So in the end we have the solution:

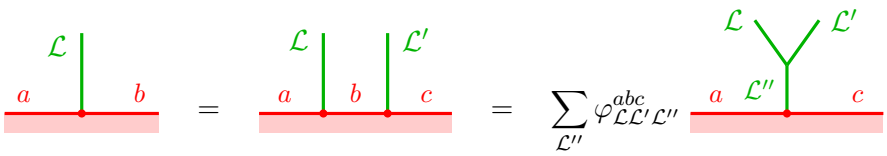
$$A = 1/\phi, \quad B = C = 1/\sqrt{\phi}, \quad D = -1/\phi. \tag{2.15}$$

In general, fixing the gauge completely is not straightforward. However, it has been proven that, up to gauge transformations, the solutions of the pentagon equations do not contain continuous parameters, as a consequence of the so-called *Ocneanu rigidity* theorem [5]. This is very important in the study of RG flows and of the preservation of F-symbols along them [4]. In the next section the F-symbols will be used with a different convention, named *tetrahedral symbols*.

$$\begin{bmatrix} \mathcal{L}_1 & \mathcal{L}_2 & \mathcal{L}_3 \\ \mathcal{L}_4 & \mathcal{L}_5 & \mathcal{L}_6 \end{bmatrix} = \frac{1}{\sqrt{d_{\mathcal{L}_3} d_{\mathcal{L}_6}}} [F^{\mathcal{L}_1\mathcal{L}_2\mathcal{L}_4}_{\mathcal{L}_5}]_{\mathcal{L}_3\mathcal{L}_6}. \tag{2.16}$$

2.3 Boundaries and φ-symbols

So far, we have seen the behaviour of the topological lines in the bulk. Next, we are interested in what happens when they interact with a boundary of the theory. Considering a set of boundary states, we expect that the transverse insertion of a topological line can change the state [1].



(2.17)

Figure 10: Fusion boundary rules.

Following the line of thought of the last section, we now want to describe the partial fusion of the lines in the boundary. This can be done using the *boundary F-symbols*. Analogously to the bulk F-symbols, the boundary F-symbols satisfy a pentagon equation

$$\sum_{\mathcal{L}_u} \left[F^{\mathcal{L}\mathcal{L}'\mathcal{L}''}_{\mathcal{L}_v}\right]_{\mathcal{L}_u\mathcal{L}'_u} \varphi^{abc}_{\mathcal{L}\mathcal{L}'\mathcal{L}_u} \varphi^{acd}_{\mathcal{L}_u\mathcal{L}''\mathcal{L}_v} = \varphi^{abd}_{\mathcal{L}\mathcal{L}'_u\mathcal{L}_v} \varphi^{bcd}_{\mathcal{L}'\mathcal{L}''\mathcal{L}'_u} \tag{2.18}$$

[1] For the same reason, considering the case of radial quantisation in a CFT, we say that the line *twists* the Hilbert space on the circle.

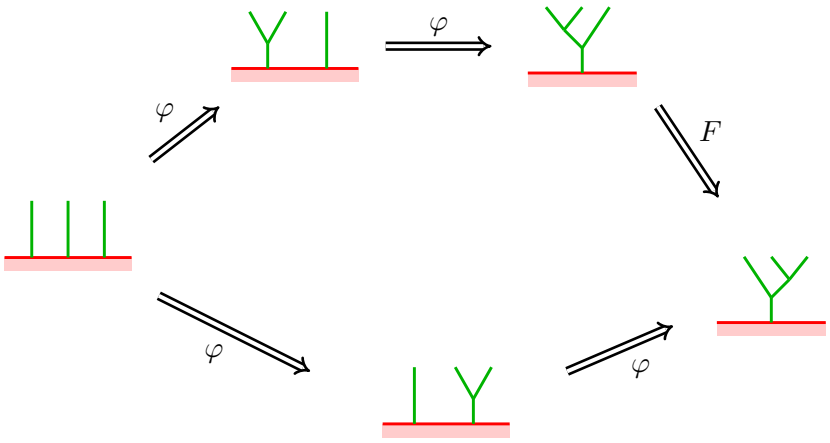
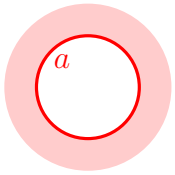


Figure 11: Boundary pentagon equation.

The information of the fusion category together with the definition of the boundary data and the boundary junction define the *module category* $\mathcal{M}$ over $\mathcal{C}$ (see Appendix A or [5]). The counterpart of the quantum dimension for the boundary data is the *relative Euler term* g_a, given by the path integral on a vacuum disc of boundary $|a\rangle$.



$$= \quad g_a \tag{2.19}$$

Figure 12: Relative Euler term g_a.

2.4 Kink Hilbert spaces

In order to make the discussion concrete, we focus on a particular type of system, characterised by having multiple vacua in a $\mathbb{R}^{1,1}$ spacetime. At a classical field theory level, this kind of system appears when the potential has multiple minima at the same height. This makes it possible to find solutions with different values of the field at $\pm\infty$, which cannot be continuously deformed into one another. These solutions are called classical *kinks*.

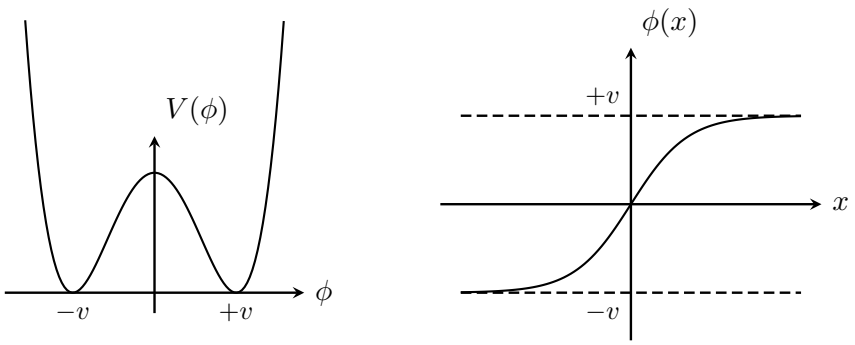


Figure 13: Example of potential with two equal minima and of kink solution.

Semiclassically, it is possible to quantize the classical solution, finding the excitations around

the classical background. At a pure quantum level, a state in a Hilbert space $|K\rangle$ is a kink if the expectation value of a local operator at $\pm\infty$ is the same as the expectation value in the vacua, different for $\pm$ infinity.

$$\lim_{x\to\pm\infty} \langle K|O(x)|K\rangle = \langle \Omega_\pm|O(0)|\Omega_\pm\rangle \,. \tag{2.20}$$

A kink with the same vacuum state at $\pm\infty$ is called a *breather*, and breathers are usually considered bound states of two kinks with opposite vacuum configuration. Since it costs infinite energy at infinity, different vacua allow us to decompose the Hilbert space into sectors that differ by the values of the vacua at $\pm\infty$.

$$\mathcal{H} = \bigoplus_{a,b} \mathcal{H}_{a,b}, \tag{2.21}$$

such that the Hamiltonian acts independently for each sector $H_{a,b} : \mathcal{H}_{a,b} \to \mathcal{H}_{a,b}$.

Adding a fusion category to the kink Hilbert space, lines that act transversely mix different Hilbert sectors:

$$[\mathcal{L}]^{cd}_{ab} : \mathcal{H}_{a,b} \to \mathcal{H}_{c,d}, \quad \mathcal{L}\,|K_{a,b}\rangle = \sum_{K_{c,d}} \sqrt{d_{\mathcal{L}}}[\mathcal{L}]^{cd}_{ab}\,|K_{c,d}\rangle \,. \tag{2.22}$$

Here, $\sqrt{d_{\mathcal{L}}}$ is introduced by convention.

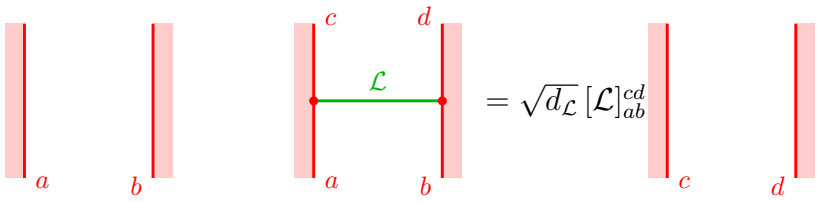


Figure 14: Boundary and action of a symmetry line on this state.

The fact that the lines are symmetries implies the commutation rule $[\mathcal{L}]^{cd}_{ab}\mathcal{H}_{a,b} = \mathcal{H}_{c,d}[\mathcal{L}]^{cd}_{ab}$. With this argument one can find that nontrivial eigenstates related by the line action $\mathcal{L}$ are degenerate in mass.

Composition of symmetry lines together with the boundary pentagon equation define the algebra of the representation of $\mathcal{L}$ on the Hilbert space.

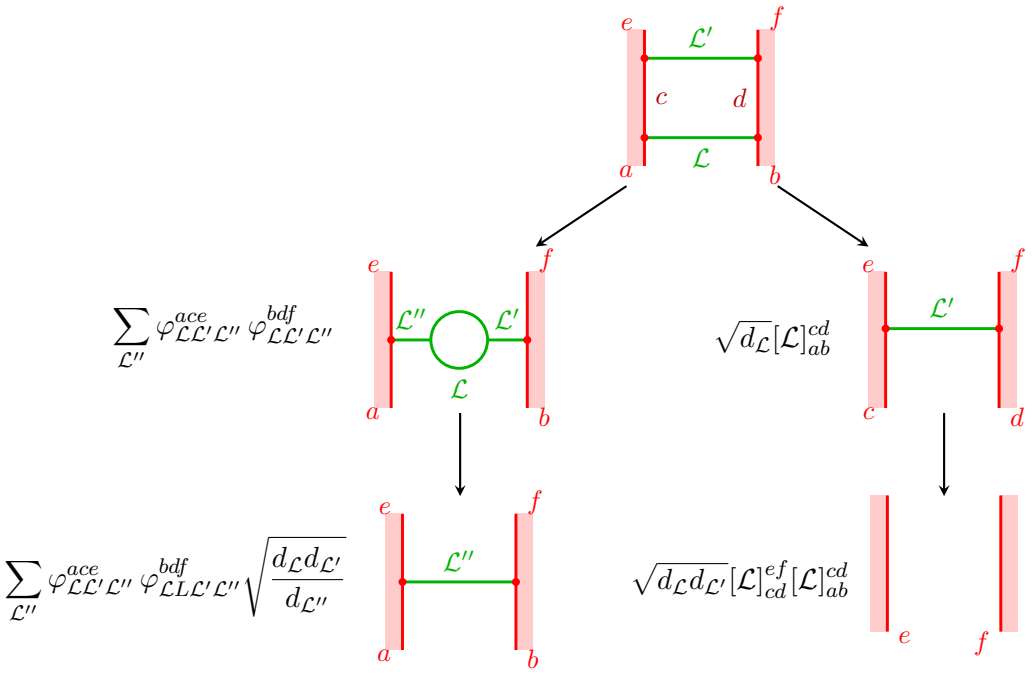


Figure 15: Diagrammatic proof of constraint for the $[\mathcal{L}]$ algebra.

$$[\mathcal{L}']^{ef}_{cd}\cdot[\mathcal{L}]^{cd}_{ab}=\sum_{\mathcal{L}''}\varphi^{ace}_{\mathcal{L}\mathcal{L}'\mathcal{L}''}\,\varphi^{cdf}_{\mathcal{L}\mathcal{L}'\mathcal{L}''}\,[\mathcal{L}'']^{ef}_{ab}\,. \tag{2.23}$$

In this context, kinks $K^{v}_{a,b}$ and breathers B^{v}_{a} are irreps of this line algebra.

2.5 Dual category and representation of kinks

In this section we develop a formalism for studying and classifying the irreducible representations (irreps) of the non-invertible line algebra. On the other side of the boundary, we define a *dual module category* $\mathcal{C}^*_{\mathcal{M}}$, see Appendix A, with the boundary φ-symbols $(\varphi^*)^{abc}_{vv'v''}$. The lines of $\mathcal{C}$ and $\mathcal{C}^*$ can be made to commute by introducing the symbols $[\mathcal{L},v]^{cd}_{ab}$.

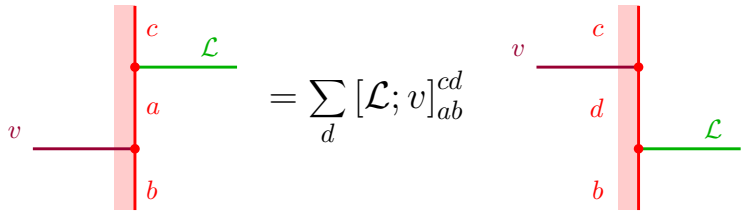


Figure 16: Commutation relation between a line in $\mathcal{C}$ and one in its dual $\mathcal{C}^*$.

Using the relation in Figure 17,

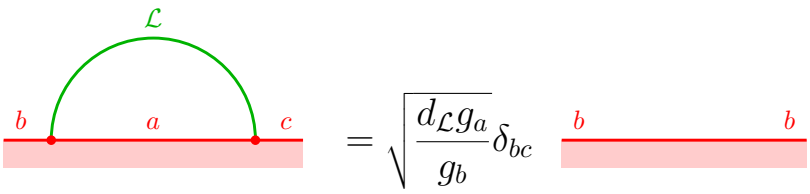


Figure 17: Contraction of a line inserted at both extremes in the boundary.

a line $\mathcal{L}$ attached to both sides of v can be simplified as in Figure 18

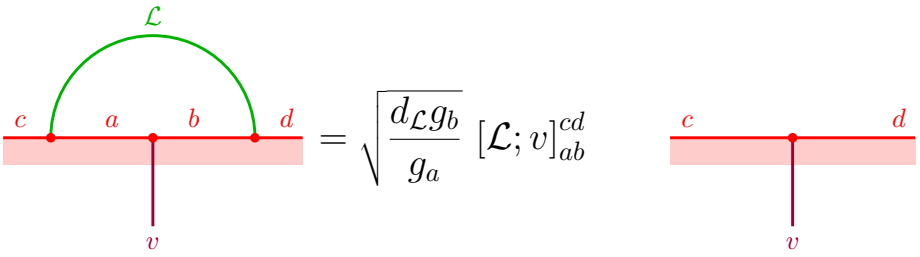


Figure 18: Contraction of $\mathcal{L}$ line closed at two sides of a v line.

Fusing two different lines $\mathcal{L}$ and $\mathcal{L}'$, one gets the same algebra obtained for $[\mathcal{L}]^{cd}_{ab}$. This is reasonable, since the kink boundary space can be, roughly speaking, obtained by deforming the space around v.

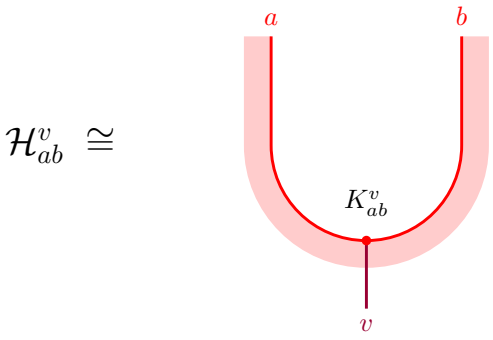


Figure 19: Isomorphism between $\mathcal{H}^v_{ab}$ and the configuration shown in figure, with a two boundary separated by a dual line v.

The non-trivial fact is that these dual lines are in one-to-one correspondence with the irreps of $[\mathcal{L}]^{cd}_{ab}$ [6]. This adds some additional structure and constraints on the representation, arising from the application of the fusion rules and pentagon equation of the dual category [3]. Given the equivalence, we can write the action of kinks on lines as:

$$\mathcal{L} \cdot K_{a,b} = \sum_{K_{c,d}} \sqrt{\frac{d_{\mathcal{L}} g_b}{g_d}} [\mathcal{L}, v]^{cd}_{ab} K_{c,d}. \tag{2.24}$$

Here, K_{ab} are the kink states, with vacua at $\pm\infty$ equal to a and b.

2.6 A second explicit example: deformed tricritical Ising regular representation

Compared with the first example, involving the Fibonacci category, we now have many more structures to determine: the boundary φ-symbols, the number of vacua and the dual category.

In general this is quite difficult, but there is a particular case that is straightforward. Given a discrete fusion category, the *regular module category* is the module category in which the symmetry lines spontaneously break the vacuum degeneracy, so that there is a bijection between the vacuum states and the symmetry lines. In particular one can imagine boundary conditions (Figure 10) labelled by the fusion lines, that are generated by juxtaposing the corresponding line operator along a "trivial boundary", in correspondence to $\mathbb{I}$ [2]. This interpretation is supported by the observation that, from Figure 11, the boundary φ-symbols can consistently be chosen to equal the bulk F-symbols. Also, we notice from Figure 16 that the dual category can be identified with the fusion category, so that the boundary becomes trivial (i.e. just a topological line of the fusion category). In this sense we have:

$$[\mathcal{L}, v]^{cd}_{ab} = \sqrt{d_a d_d} \begin{bmatrix} \mathcal{L} & c & a \\ v & b & d \end{bmatrix} . \tag{2.25}$$

A legitimate question to ask at this point is: does there exist an interesting model with this kind of representation? Luckily, the answer is positive, and is given by some deformations induced by primary fields of the minimal models $\mathcal{M}_{n+1,n}$. One can show that each $\mathcal{M}_{n+1,n}$ has a set of topological lines, called *Verlinde lines* [7], in bijection with the primary operators of the CFT. Furthermore, under a deformation of the theory by the primary field $\phi_{1,3}$, $n-1$ lines are preserved, forming the $\mathcal{A}_n$ fusion category.

$$\mathcal{A}_n = \{\mathcal{L}_{r,1}, r = 1, .., n-1\} . \tag{2.26}$$

Labelling the lines with $a = 0, 1/2, .., n/2 - 1$, they satisfy the fusion rule:

$$a \times b = \sum_{c=|a-b|,\ c-a+b=0(mod 1)}^{min(a+b, n-2+a+b)} c. \tag{2.27}$$

In these deformed theories, it is possible to spontaneously break the symmetry completely, obtaining the regular module category and the regular representation. The simplest example of these minimal models is the tricritical Ising model $\mathcal{M}_{5,4}$, described in [8]. Its Verlinde lines are summarized in the table. Starting from this minimal model, there are two deformations that

Tricritical Ising Model $\mathcal{M}_{5,4}$			
Kac label	Conformal Weight	Verlinde Line	Quantum Dimension
$(1,1)$ or $(3,4)$	$h_{1,1} = 0$	1	$d_{1,1} = 1$
$(1,2)$ or $(3,3)$	$h_{1,2} = 1/10$	$W \times \eta$	$d_{1,2} = \dfrac{1+\sqrt{5}}{2}$
$(1,3)$ or $(3,2)$	$h_{1,3} = 3/5$	W	$d_{1,3} = \dfrac{1+\sqrt{5}}{2}$
$(1,4)$ or $(3,1)$	$h_{1,4} = 3/2$	η	$d_{1,4} = 1$
$(2,2)$ or $(2,3)$	$h_{2,2} = 3/80$	$W \times N$	$d_{2,2} = \sqrt{2}\left(\dfrac{1+\sqrt{5}}{2}\right)$
$(2,4)$ or $(2,1)$	$h_{2,1} = 7/16$	N	$d_{2,1} = \sqrt{2}$

preserve a fusion-category symmetry and feature a module category in the regular representation.

[2] Observe that this is similar to the way traditional symmetries that come from Noether's theorem are defined $\int \mathrm{d}x j_0(x)$, where the integration is performed along a spacelike hypersurface.

The $\phi_{1,2}$ deformation, which preserves the Fibonacci category, with $\{1, W\}$, two minima $1, W$ and $v = W$, and define two kinks and one breather $K^W_{1,W}, K^W_{W,1}, B^W_W$, all with the same mass.

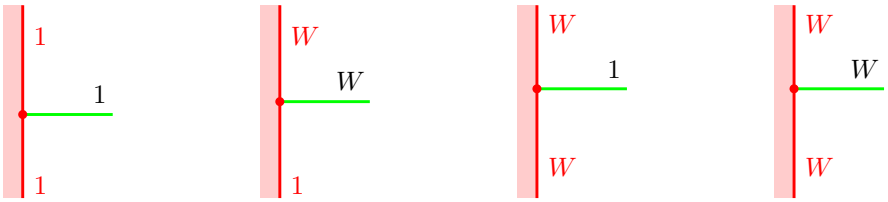


Figure 20: Possible boundary configurations of $\phi_{1,2}$ deformation of tricritical Ising.

The second deformation is the previously discussed $\phi_{1,3}$ deformation, which preserves the $\mathbb{Z}_2$ *Tambara-Yamagami fusion category*:

$$\eta \times \eta = 1, \quad N \times N = 1 + \eta, \quad N \times \eta = \eta \times N = N. \tag{2.28}$$

The deformed theory has three distinct vacua. Contrary to the $\phi_{1,2}$ deformation, in this case there are no kink bound states.

3. Constraints on S-matrix: modifying crossing symmetries

In the second chapter we introduced the formalism of fusion categories and their representations in a 1+1D QFT. In this section we apply this knowledge in order to gain some insight into the dynamics, in particular to the S-matrix of the theory. Deriving the exact S-matrix of a theory is usually difficult, and is possible in very few cases. The main approach to attack this problem is the so-called *S-matrix bootstrap*. Details of the S-matrix bootstrap are reviewed in Appendix C and in [9, 10]. In the bootstrap approach there is a fundamental assumption that one may impose, namely the *Yang–Baxter equation*. With this assumption one can develop the *integrable S-matrix bootstrap*, which has made it possible to obtain numerous exact analytical results in the last decades. Without this assumption, it is possible to use numerical techniques, applying the *numerical bootstrap*. In this chapter, following [2], we restrict ourselves to integrable models with regular representation describing the symmetry category, in particular to $\phi_{1,3}$ deformations of $\mathcal{M}_n$ minimal models (which are indeed integrable).

3.1 ZF algebra for kinks

We start by defining the ZF algebra in theories where the fundamental excitations are kinks. Defining K_{ab} a kink with vacua labeled by a, b, two kinks can scatter only if the $-\infty$ vacuum of one kink is the $+\infty$ vacuum of the other. Each kink has a momentum p_i and an energy E_i, that can be written in terms of the rapidity variable θ by the relations $p_i = m_i \sinh(\theta_i)$, $E_i = m_i \cosh(\theta_i)$. Using the fact that under boosts the rapidities are translated by a constant, and that in a relativistic theory the S-matrix is Lorentz invariant, we expect the S-matrix to depend only on the combination $\theta_{12} := \theta_1 - \theta_2$. Furthermore, by the conservation of topological charge, we require that the types of vacua at infinity do not change during the scattering. With these two requirements we can obtain the following ZF algebra.

$$K_{dc}(\theta_1)K_{cb}(\theta_2) = \sum_a S^{ab}_{dc}(\theta_{12})K_{da}(\theta_2)K_{ab}(\theta_1), \quad \theta_{12} > 0. \tag{3.1}$$

$S^{ab}_{dc}(\theta_{12}) =$

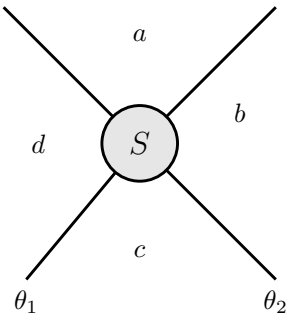


Figure 21: Diagrammatic representation of S-matrix $S^{ab}_{dc}(\theta_{12})$.

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3.2 Classical crossing relation and S-matrix

In order to fix the overall function in front of the S-matrix, it is necessary to impose the *crossing equation*. The idea is to consider a particular vacuum state, obtained as bound state with zero

energy, momentum and higher order charges. The general ansatz is of the form:

$$\Omega_A(\theta) = \sum_{a\in A} \lambda_a \mathbb{A}_a^\dagger(\theta)\mathbb{A}_a^\dagger(\theta + i\pi). \tag{3.2}$$

Here, $\mathbb{A}_a^\dagger(\theta)$ are general elements of the ZF algebra (in our case they correspond to $K_{ab}(\theta)$), and the vanishing condition of energy and momentum are verified by $P\mathbb{A}_a^\dagger(\theta)\mathbb{A}_a^\dagger(\theta+i\pi)\left|0\right\rangle = (p(\theta)+p(\theta+i\pi))\mathbb{A}_a^\dagger(\theta)\mathbb{A}_a^\dagger(\theta+i\pi)\left|0\right\rangle = m(\sinh(\theta)+\sinh(\theta+i\pi))\mathbb{A}_a^\dagger(\theta)\mathbb{A}_a^\dagger(\theta+i\pi)\left|0\right\rangle = 0$.

We also require that it commutes with every $\mathbb{A}_a^\dagger$ and is a singlet under any symmetry of the theory. For example, in the case of a purely elastic theory, i.e. with S-matrix in the form $S_{ab}^{cd}(\theta) = S(\theta)\delta_a^c\delta_b^d$ with braiding unitarity we obtain:

$$S(\theta) = S(i\pi - \theta). \tag{3.3}$$

In general, the more general equation is assumed:

$$S_{ij}^{kl}(\theta) = S_{j\bar{k}}^{l\bar{i}}(i\pi - \theta). \tag{3.4}$$

In the case of kinks, last equation translates into:

$$S_{dc}^{ab}(\theta) = S_{ad}^{bc}(i\pi - \theta). \tag{3.5}$$

Focusing on the tricritical Ising model, one can show that the correct vacuum state is given by:

$$\Omega(\theta) = \Omega_1(\theta) + \Omega_2(\theta) + \Omega_3(\theta) + \Omega_4(\theta). \tag{3.6}$$

where:

$$\begin{aligned} &\Omega_1(\theta) = K_{0\frac{1}{2}}(\theta)K_{\frac{1}{2}0}(\theta+i\pi), \quad \Omega_2(\theta) = K_{\frac{1}{2}0}(\theta)K_{0\frac{1}{2}}(\theta+i\pi), \\ &\Omega_3(\theta) = K_{\frac{1}{2}1}(\theta)K_{1\frac{1}{2}}(\theta+i\pi), \quad \Omega_4(\theta) = K_{1\frac{1}{2}}(\theta)K_{\frac{1}{2}1}(\theta+i\pi). \end{aligned} \tag{3.7}$$

By applying integrable S-matrix bootstrap, with crossing equation (3.5) one can derive a candidate for the S-matrix for the $\phi_{1,3}$ deformed $\mathcal{M}_n$ minimal models (see Appendix C) for a general introduction and [11] for a derivation):

$$S_{dc}^{ab}(\theta) = Z(\theta)\left(\frac{d_a d_c}{d_b d_d}\right)^{\frac{i\theta}{2\pi}}\left[\sqrt{\frac{d_a d_c}{d_b d_d}}\sinh\left(\frac{\theta}{n}\right)\delta_{bd} + \sinh\left(\frac{i\pi-\theta}{n}\right)\delta_{ac}\right] \tag{3.8}$$

Here, $Z(\theta) = \frac{1}{\sinh\frac{\theta-i\pi}{n}}\exp\left\{\frac{i}{2}\int_{-\infty}^{\infty}\frac{dk}{k}\sin k\theta\,\frac{\sinh\frac{k\pi}{2}(n-1)}{\sinh\frac{nk\pi}{2}\cosh\frac{k\pi}{2}}\right\}$ and $d_a = \frac{sin(\pi(2a+1)/n)}{\pi/n}$ are the quantum dimensions of the different lines. In the next section we will see that this ansatz is incompatible with Ward identities of non-invertible symmetries.

3.3 Ward identities and inconsistencies

Since the non-invertible lines commute with the stress-energy tensor, they also commute with the S-matrix. The action of symmetry lines on kinks is given by (2.24). In the regular representation, the $[\mathcal{L}, v]$ symbols can be written using (2.25). For the case of $\mathcal{A}_4$ category corresponding to $\phi_{1,3}$ deformed tricritical Ising, the explicit tetrahedral symbols are given in Appendix B of [3].

Remembering the correspondence between vacua states and fusion lines, we can expect that their action generates kinks compatible with fusion rules.

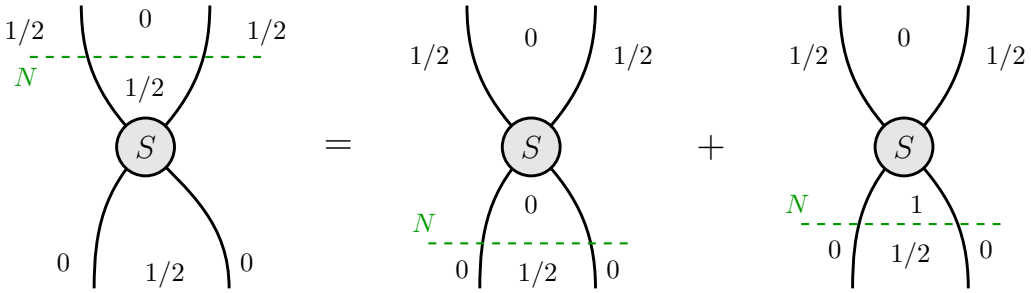


Figure 22: Diagrammatic form of commutation relation between N line and S-matrix.

For example, the action of the N line on the $K_{1/2,0}$ kink state is equivalent to considering:

$$\mathcal{L}_N \cdot K_{1/2\ 0} := [N,N]K_{1/2\ 0} = a_1 K_{0\,1/2} + a_2 K_{1\,1/2}. \tag{3.9}$$

Notice that we used the fact that, for the $\mathcal{A}_n$ category coming from deformed minimal models, $v = 1/2 = N$ (see [3]). The other terms on the right-hand side are excluded by the fusion rules, and the coefficients a_1, a_2 are determined by F-symbols.

In order to gain information about the S-matrix we use the fact that $\mathcal{L}_N$ and the S-matrix commute. Starting from $K_{0\,\frac{1}{2}}(\theta_1)K_{\frac{1}{2}\,0}(\theta_2)$ we can either apply the $\mathcal{L}_N$ line first and then use the ZF algebra, as done in (3.10), or use the ZF algebra first and then apply the line, as in (3.11).

$$\begin{aligned}
&\mathcal{L}_N \cdot K_{0\,\frac{1}{2}}(\theta_1)K_{\frac{1}{2}\,0}(\theta_2) \\
&= K_{\frac{1}{2}\,0}(\theta_1)K_{0\,\frac{1}{2}}(\theta_2) + K_{\frac{1}{2}\,1}(\theta_1)K_{1\,\frac{1}{2}}(\theta_2) \\
&= K_{\frac{1}{2}\,0}(\theta_2)K_{0\,\frac{1}{2}}(\theta_1)S_{\frac{1}{2}\,0}^{0\,\frac{1}{2}}(\theta) + K_{\frac{1}{2}\,1}(\theta_2)K_{1\,\frac{1}{2}}(\theta_1)S_{\frac{1}{2}\,0}^{1\,\frac{1}{2}}(\theta) \\
&\quad + K_{\frac{1}{2}\,1}(\theta_2)K_{1\,\frac{1}{2}}(\theta_1)S_{\frac{1}{2}\,1}^{1\,\frac{1}{2}}(\theta) + K_{\frac{1}{2}\,0}(\theta_2)K_{0\,\frac{1}{2}}(\theta_1)S_{\frac{1}{2}\,1}^{0\,\frac{1}{2}}(\theta).
\end{aligned} \tag{3.10}$$

$$\begin{aligned}
&\mathcal{L}_N \cdot K_{0\,\frac{1}{2}}(\theta_1)K_{\frac{1}{2}\,0}(\theta_2) \\
&= \mathcal{L}_N \cdot K_{0\,\frac{1}{2}}(\theta_2)K_{\frac{1}{2}\,0}(\theta_1)S_{0\,\frac{1}{2}}^{\frac{1}{2}\,0}(\theta) \\
&= K_{\frac{1}{2}\,0}(\theta_2)K_{0\,\frac{1}{2}}(\theta_1)S_{0\,\frac{1}{2}}^{\frac{1}{2}\,0}(\theta) + K_{\frac{1}{2}\,1}(\theta_2)K_{1\,\frac{1}{2}}(\theta_1)S_{0\,\frac{1}{2}}^{\frac{1}{2}\,0}(\theta).
\end{aligned} \tag{3.11}$$

Equating the coefficients of the KK terms, we arrive at the Ward identities:

$$\begin{aligned}
S_{0\,\frac{1}{2}}^{\frac{1}{2}\,0}(\theta) &= S_{\frac{1}{2}\,0}^{0\,\frac{1}{2}}(\theta) + S_{\frac{1}{2}\,1}^{0\,\frac{1}{2}}(\theta), \\
S_{0\,\frac{1}{2}}^{\frac{1}{2}\,0}(\theta) &= S_{\frac{1}{2}\,0}^{1\,\frac{1}{2}}(\theta) + S_{\frac{1}{2}\,1}^{1\,\frac{1}{2}}(\theta).
\end{aligned} \tag{3.12}$$

A schematic picture of the first identity of (3.12) is given in Figure 22. We can now check if the ansatz (3.8) verifies this relation. Inserting the explicit value of the S-matrix we obtain:

$$\left(\sqrt{2}\,\sinh\frac{\theta}{4} + \sinh\left(\frac{i\pi-\theta}{4}\right)\right)\sqrt{2}^{\frac{i\theta}{\pi}} \neq \left(\frac{1}{\sqrt{2}}\,\sinh\frac{\theta}{4} + \sinh\left(\frac{i\pi-\theta}{4}\right) + \frac{1}{\sqrt{2}}\,\sinh\frac{\theta}{4}\right)\left(\frac{1}{\sqrt{2}}\right)^{\frac{i\theta}{\pi}}. \tag{3.13}$$

This implies that the initial ansatz for the S-matrix was incorrect.

3.4 Modifying crossing

In this section we sketch an argument in order to derive the modified crossing relation. Usually, S-matrix and scattering amplitudes are obtained by correlation functions using the LSZ formula. However, this is true for asymptotically free local fields, a requirement that kinks do not satisfy. Nevertheless, since their energy profiles are localised, we may hypothesise that the S-matrix is proportional to the analytic continuation of a correlation function of insertions of kinks on a large disc, and we consider this part actually crossing symmetric (in the standard sense). A discussion of this point is given in [12]. What actually changes when the channel is changed is the normalisation of the IN and OUT states. Figure 23 shows a diagrammatic representation of these states. For example, the IN diagram represents two kinks infinitely far apart, the first starts at $-\infty$ with value d and assumes the value c in the bulk, which is also the initial value of the second kink, whose value at $+\infty$ is b. Since the S-matrix has to be unitary, the asymptotic states must be normalised; consequently, the correlation function is divided by an overall factor $\sqrt{\langle IN|IN\rangle \langle OUT|OUT\rangle}$. The diagrammatic representation of this is the third part of Figure 23.

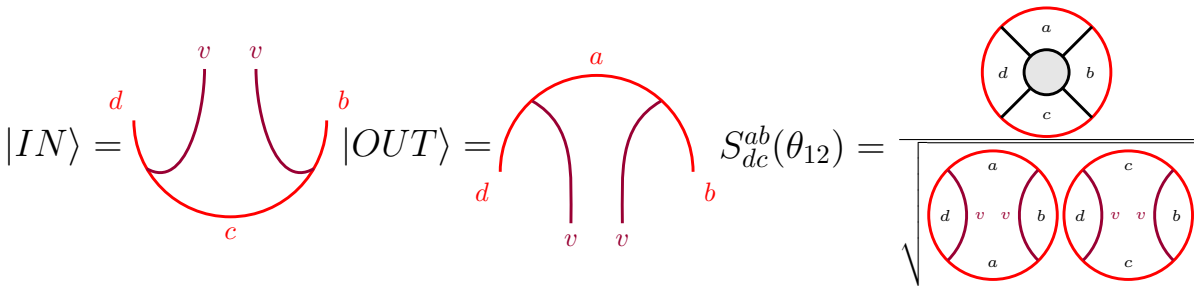


Figure 23: IN and OUT states, and schematic picture of S-matrix, written as the ratio between the euclidean correlation function and the square of the product of the normalization of IN and OUT states.

The normalisation can be computed by successively applying the orthogonality relation (Figure 3) and the quantum-dimension relation (Figure 4). For the IN state, the calculation is shown in Figure 24.

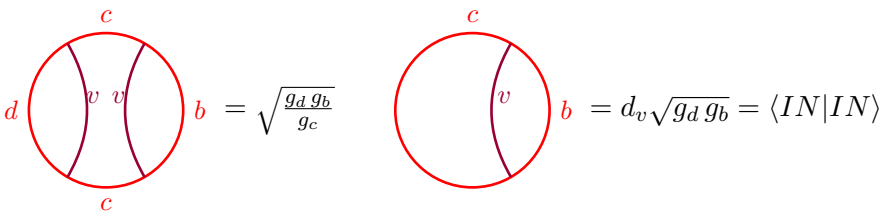


Figure 24: Diagrammatic computation of IN normalisation.

Having computed the normalisations in the regular representation, by comparing the s-channel and the t-channel correlation functions (the IN and OUT diagrams can be found by simply observing Figure 21), we obtain the equation:

$$\sqrt{d_b d_d} S^{ab}_{dc}(s) = \sqrt{d_a d_c} S^{bc}_{ad}(t). \tag{3.14}$$

Using that, in 2D for kinks with degenerate masses, $t = 4m^2 - s$, we arrive at the modified crossing equation (3.15).

$$S_{dc}^{ab}(s) = \sqrt{\frac{d_a d_c}{d_b d_d}} S_{ad}^{bc}(4m^2 - s). \tag{3.15}$$

With this new crossing relation we can try to find the new S-matrix via the integrable bootstrap. The new S-matrix is:

$$S_{dc}^{ab}(\theta) = Z(\theta) \left[\sqrt{\frac{d_a d_c}{d_b d_d}} \sinh\left(\frac{\theta}{n}\right) \delta_{bd} + \sinh\left(\frac{i\pi - \theta}{n}\right) \delta_{ac} \right]. \tag{3.16}$$

It differs from the previous ansatz by the absence of the $\left(\frac{d_a d_c}{d_b d_d}\right)^{\frac{i\theta}{2\pi}}$ factor.

Again, we can verify the Ward identity, obtaining:

$$\sqrt{2} \sinh\frac{\theta}{4} + \sinh\left(\frac{i\pi - \theta}{4}\right) = \frac{1}{\sqrt{2}} \sinh\frac{\theta}{4} + \sinh\left(\frac{i\pi - \theta}{4}\right) + \frac{1}{\sqrt{2}} \sinh\frac{\theta}{4}. \tag{3.17}$$

Which is now a true identity.

4. SymTFT and numerical bootstrap

In this final chapter we use, at an operational level, the SymTFT formalism in order to study systematically the numerical S-matrix bootstrap for systems with non-invertible symmetries. The operative rules for the representation of module categories, dual categories and S-matrix can be found in D, whereas for the mathematical details, see [8, 13]. In particular, using this formalism it is possible to find for a general representation of the module category the equation for modified crossing:

$$S_{ab}^{dc}(s) \;=\; \sqrt{\frac{g_a g_c}{g_b g_d}}\; S_{ad}^{bc}(t) \tag{4.1}$$

Where, in comparison to (3.15) we have the Euler relative terms g_i instead of the quantum dimensions d_i. Also, the TFT formalism allows us to generalize Ward identities (3.12), writing for a general representation if $[\mathcal{L}; v]$ symbols:

$$\sum_e [\mathcal{L}; v]_{d'c'}^{de} [\mathcal{L}; v]_{b'c'}^{be} \sqrt{\frac{g_c g_b g_d}{g_e}}\, S_{de}^{ab}(s) = \sum_{e'} [\mathcal{L}; v]_{da}^{d'e'} [\mathcal{L}; v]_{ba}^{b'e'} \sqrt{\frac{g_a g_{b'} g_{d'}}{g_{e'}}}\, S_{d'c'}^{e'b'}(s). \tag{4.2}$$

4.1 Projector Basis formalism

In order to write an S-matrix that satisfy the general form of the Ward identities (4.2), we decompose the S-matrix into a sum of projectors.

$$S_{dc}^{ab}(s) = \sum_{\chi \in v \times v} (P_\chi)_{dc}^{ab} A_\chi(s). \tag{4.3}$$

In the SymTFT context this corresponds to inserting a completeness relation in the bulk, as in Figure 25.

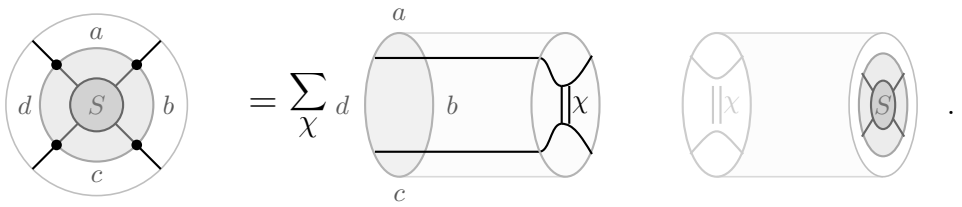


Figure 25: TFT representation of insertion of a complete set of states.

where:

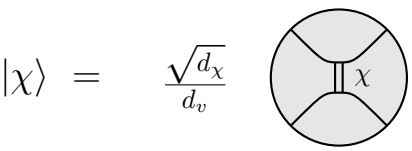


Figure 26: Diagrammatic definition of projector state.

In equation (4.3), the left-hand side corresponds to the S-matrix, whereas the right-hand side contains the projectors P_χ and the amplitudes $A_\chi(s)$. Projectors can be evaluated in terms of φ^* symbols, obtaining the RHS of Figure 27:

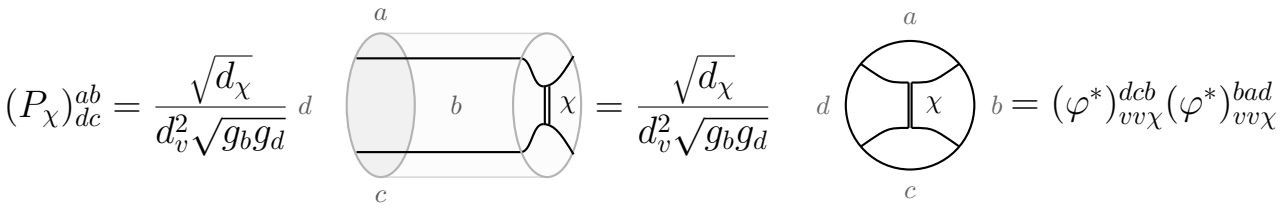


Figure 27: TFT computation of the projector.

For example, in the regular representation, this gives:

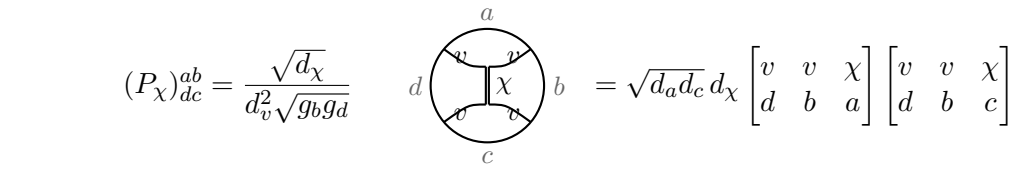


Figure 28: Expression of projectors in regular representation.

Diagrammatically, we see that projectors trivially solve the Ward identities.

4.2 Non-invertible symmetries and numerical bootstrap

Finally, once the S-matrix is written in terms of the projectors, we can easily incorporate non-invertible symmetries in S-matrix bootstrap. Below, we describe how each property imposed in the bootstrap is implemented.

1. Unitarity condition takes the form:

$$\sum_e S^{eb}_{dc}(s) S^{ab}_{de}(s^*) \leq \delta_{ac}. \tag{4.4}$$

   The inequality $\leq$ arises from the fact that for each IN state we are neglecting multiple OUT states. Using orthogonality of projectors this becomes:

$$A_\chi(s) A_\chi(s^*) \leq 1, \quad \forall \chi, \forall s \geq 4m^2. \tag{4.5}$$

2. Modified crossing equation (3.15) mixes the dependence of different $A_\chi(s)$. We can perform the computation leading to equation (4.6),

$$= \sum_\chi A_\chi(s) \frac{\sqrt{d_\chi}}{d_v} \quad = \sum_{\chi'} A_{\chi'}(4m^2 - s) \frac{\sqrt{d_{\chi'}}}{d_v} \tag{4.6}$$

$$= \sum_{\chi,\chi'} \frac{\sqrt{d_{\chi'}}}{d_v} [F^v_{vvv}]^{\chi\chi'}_* A_{\chi'}(4m^2 - s) \tag{4.7}$$

$$= \sum_{\chi,\chi'} \frac{d_{\chi'}\sqrt{d_\chi}}{d_v} \begin{bmatrix} v & v & \chi \\ v & v & \chi' \end{bmatrix}_* A_{\chi'}(4m^2 - s) \tag{4.8}$$

arriving at the formula:

$$A_\chi(s) = \sum_{\chi'} C_{\chi\chi'} A_{\chi'}(4m^2 - s), \quad C_{\chi\chi'} = d_{\chi'} \begin{bmatrix} v & v & \chi \\ v & v & \chi' \end{bmatrix}_* . \tag{4.9}$$

3. Analyticity is implemented by requiring that the amplitudes $A_\chi(s)$ are analytic functions in the physical sheet, except for possible bound states between 0 and the multiparticle threshold $4m^2$.

4. Numerical bootstrap for the amplitude at a point s_* can be performed by writing an ansatz for $A_\chi(s)$ that satisfies crossing and analyticity and that depends on a finite set of parameters, constraining this ansatz using unitarity $|A_\chi(s_j)|^2 \leq 1$ for different choices of $s_j \geq 4m^2$, and finally finding the boundary of the allowed region for $A_\chi(s_*)$ by extremising the functional $\mathcal{F} = \sum_\chi n_\chi A_\chi(s_*)$ for different fixed values of n_χ.

5. In order to verify whether some of the points satisfy the integrability condition, we can formulate the Yang–Baxter equation in terms of the amplitudes and, by applying a series of F-moves, derive:

$$\begin{aligned} &\sum_{\eta_2} d_{\eta_2} \begin{bmatrix} v & v & \eta_2 \\ \xi_2 & v & \eta_1 \end{bmatrix}_* \begin{bmatrix} v & v & \eta_2 \\ \xi_2 & v & \eta_3 \end{bmatrix}_* A_{\eta_1}(\theta_2) A_{\eta_2}(\theta_1 + \theta_2) A_{\eta_3}(\theta_1) = \\ &\sum_{\chi_1,\chi_2,\chi_3} d_{\chi_1} d_{\chi_2} d_{\chi_3} \begin{bmatrix} v & v & \chi_2 \\ \xi_2 & v & \chi_1 \end{bmatrix}_* \begin{bmatrix} v & v & \chi_2 \\ \xi_2 & v & \chi_3 \end{bmatrix}_* \begin{bmatrix} v & v & \chi_1 \\ \xi_2 & v & \eta_1 \end{bmatrix}_* \begin{bmatrix} v & v & \chi_3 \\ \xi_2 & v & \eta_3 \end{bmatrix}_* A_{\chi_1}(\theta_1) A_{\chi_2}(\theta_1 + \theta_2) A_{\chi_3}(\theta_2). \end{aligned} \tag{4.10}$$

4.3 Final explicit example: bootstrap for the $\mathcal{A}_n$ and Fibonacci categories

Finally, we make the discussion concrete by considering two distinct examples of numerical S-matrix bootstrap with a fusion category: the $\mathcal{A}_n$ category and the Fibonacci category. In the case of the $\mathcal{A}_n$ category, for the $v = \frac{1}{2}$ (regular) representation we have two possible fusion channels $\chi = 0, 1$. The crossing relation takes the explicit form:

$$\begin{pmatrix} A_0(s) \\ A_1(s) \end{pmatrix} = \frac{1}{\mathrm{d}_{1/2}} \begin{pmatrix} 1 & \mathrm{d}_1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} A_0(4m^2 - s) \\ A_1(4m^2 - s) \end{pmatrix}, \qquad \mathrm{d}_a = \frac{\sin(\pi(2a+1)/n)}{\sin(\pi/n)}. \tag{4.11}$$

In this case we have $v \not\subset v \times v$, so there are no bound states to consider. Knowing the analytic structure of the amplitudes, we can guess a structure of the form:

$$A_\chi(s) = \sum_{n=0}^{N_{max}} \Big[a_\chi^{(n)} \rho_{s_0}^n(s) + C_{\chi\chi'} a_{\chi'}^{(n)} \rho_{s_0}^n(4m^2 - s) \Big]. \tag{4.12}$$

where

$$\rho_{s_0}(s) := \frac{\sqrt{4m^2 - s_0} - \sqrt{4m^2 - s}}{\sqrt{4m^2 - s_0} + \sqrt{4m^2 - s}}, \tag{4.13}$$

$C_{\chi\chi'}$ is the crossing matrix, s_0 and $N_{\max}$ are free parameters to be chosen and $a_\chi{}^{(n)}$ are the parameters to be constrained. Notice that our ansatz satisfies both analyticity and crossing.

The unitarity condition can be implemented by choosing a set of $s_j \geq 4m^2$ and requiring that

$$|A_\chi(s_j)|^2 \leq 1. \tag{4.14}$$

Following [3], we choose the values of s_j as follows.

$$s_j = \frac{8m^2}{1+\cos(\phi_j)}, \quad \phi_j = \frac{\pi}{2}\Big[1 - \cos\left(\frac{2j-1}{N_{grid}}\pi\right)\Big], \quad j = 1, ..., N_{grid}. \tag{4.15}$$

Under these constraints, we extremise the functional $\mathcal{F} = n_\chi A_\chi(s_*) + n_{\chi'} A_{\chi'}(s_*)$, for 128 values of n_χ equally spaced around the circle. Plotting the results, one obtains:

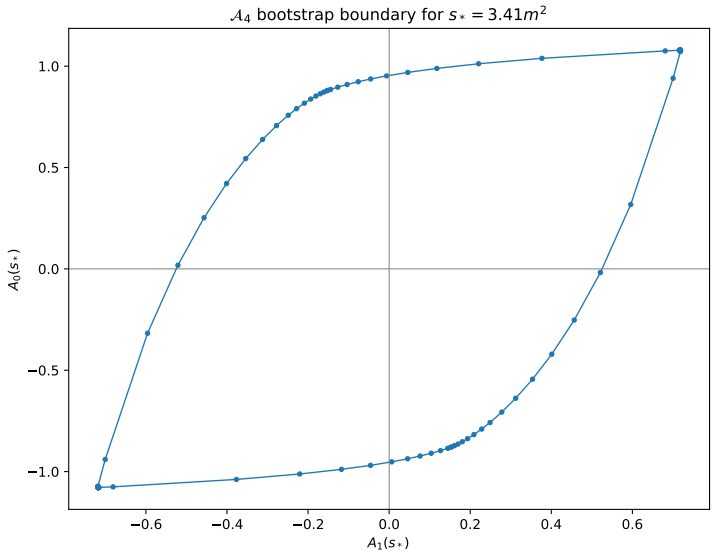


Figure 29: Points and regression curve for the boundary of the allowed-amplitude region, from the numerical bootstrap approach.

Notice that, with the chosen grid and functional, more points are obtained in regions where the boundary has greater curvature. The plot is symmetric under $A_\chi \to -A_\chi$ since the functional has the same type of symmetry. A second observation is that, for the points at the vertices of the figure, one can check that they are integrable satisfying (4.10), and in particular they are the integrable $\phi_{1,3}$ deformations of $\mathcal{M}_n$ minimal models discussed above.
The second analysis concerns the space of models with the Fibonacci category, in the $v = 1/2$ (regular) representation. In this case we have the amplitudes $A_0,\ A_{1/2}$, which satisfy the crossing relation:

$$\begin{pmatrix} A_0(s) \\ A_{1/2}(s) \end{pmatrix} = \frac{1}{\mathrm{d}_{1/2}} \begin{pmatrix} 1 & \mathrm{d}_{1/2} \\ 1 & -1 \end{pmatrix} \begin{pmatrix} A_0(4m^2 - s) \\ A_{1/2}(4m^2 - s) \end{pmatrix}, \qquad \mathrm{d}_{1/2} = \frac{1+\sqrt{5}}{2}. \tag{4.16}$$

Contrary to the $\mathcal{A}_n$ case, here we have $v \subset v \times v$, and the triple coupling can be interpreted as a bound state, in particular the expected breather $K_{1/2,1/2}$. Since it has the same mass as the kinks, we expect that:

$$A_{1/2} \approx \frac{g^2}{s - m^2}. \tag{4.17}$$

Given the bound-state data, we can take the bootstrap space to be $\{A_0(s_*), A_{1/2}(s_*), g^2\}$.

The result for $s \approx 3.41\ m^2$ is the following.

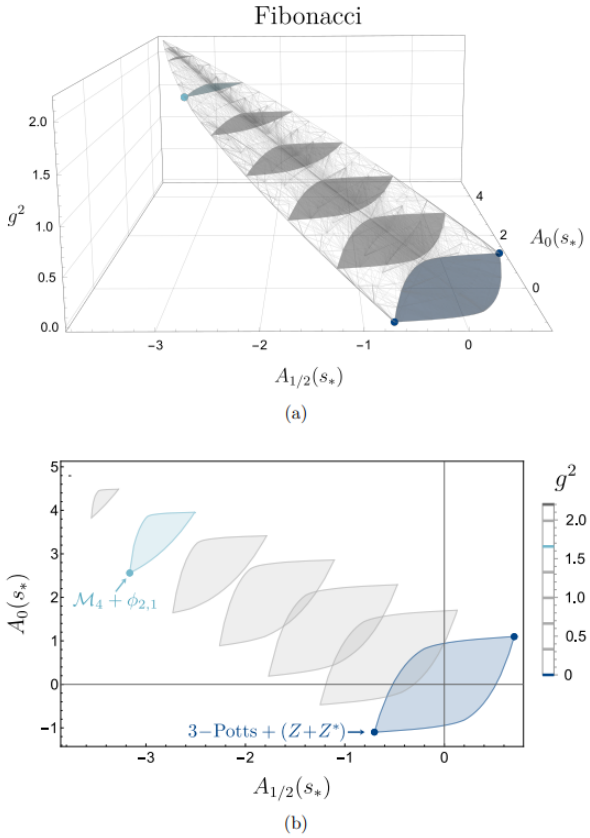


Figure 30: Numerical bootstrap of Fibonacci category. Plots taken from [3].

Note that:

- For $g^2 = 0$ we have formally the same space of $\mathcal{A}_5$, since $d_{1/2}^{\mathcal{A}_5} = d_1^{\mathcal{A}_5} = d_{1/2}^{Fibo}$. However, the theory at the vertex is different, because the number of vacua and the total preserved symmetry are different: the symmetry category is $Fib \times \mathbb{Z}_2$, rather than $\mathcal{A}_5$.
- For $g^2 \approx 2.19$ the amplitude space is a point, since there are no values of g^2 greater than this value that satisfy analyticity, crossing and unitarity.
- At a vertex of a slice with $g^2 \neq 0$ we recover the $\phi_{1,2}$ integrable deformation of the tricritical Ising model analysed before.

5. Conclusions

In this work, we review articles [2] and [3], exploring the possibility of implementing non-invertible symmetries in bootstrap programs formulated in 2D QFTs. A possible next step could be to investigate what happens when increasing the number of spacetime dimensions, studying the Ward identities associated to the S-matrix, and the restriction of the numerical bootstrap.

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A. Categories and Non-Invertible Symmetries

In this appendix we justify the use of the terminology *fusion category* for describing the set of topological lines. For the mathematical details, see [4, 5, 6, 13]. In order to define a category, we need the following elements:

1. A collection of *objects* $Obj(\mathcal{C})$;
2. For each $a, b \in Obj(\mathcal{C})$ the set of *morphisms* $Hom_{\mathcal{C}}(a, b)$, with elements $f : a \to b$. Related to the morphisms, we have a composition rule:

$$\circ : \mathrm{Hom}_{\mathcal{C}}(b, c) \times \mathrm{Hom}_{\mathcal{C}}(a, b) \to \mathrm{Hom}_{\mathcal{C}}(a, c), \qquad \forall a, b, c. \tag{A.1}$$

   The composition must satisfy associativity and an identity element should exist.

In our case, it is easy to identify the corresponding elements of this formalism.

1. Objects are given by symmetry lines $a, b, c, ... \in \mathcal{C}$.
2. Morphisms are given by topological local operators inserted between two symmetry lines $O_a^b \in Hom(a, b)$. We call an object a *simple* if $\mathrm{Hom}(a, a) \cong \mathbb{C}$.

To the category we add a fusion property:

$$a \otimes b = \sum_c N_{a\ b}^c\ c. \tag{A.2}$$

Here, the $N_{a\ b}^c$ define the dimension of the vector space on the trivalent junction $Hom(a \otimes b, c)$. Given a fusion category $\mathcal{C}$, and a category $\mathcal{M}$, the *module category* $\mathcal{M}$ over $\mathcal{C}$ is the pair $(\mathcal{M}, \mathcal{C})$ with an operation:

$$\cdot : \mathcal{C} \times \mathcal{M} \to \mathcal{M} \tag{A.3}$$

$$(a, i) \to a \cdot i \in \mathcal{M}, \quad a \in \mathcal{C}, i \in \mathcal{M}. \tag{A.4}$$

Up to isomorphism, this operation respects the associativity and the unit property of the fusion category $\mathcal{C}$.

$$(a \otimes b) \cdot i \sim a \cdot (b \cdot i), \qquad \mathbb{I} \cdot i = i. \tag{A.5}$$

In the case of the kink Hilbert space, the category $\mathcal{M}$ is given by the set of vacua of the theory, and the action of $\mathcal{C}$ is pictorially described by Figure 10. In analogy with the fusion category, associativity property of module category allows to define the boundary F-symbols φ described in the main text, with their own boundary pentagon equation.
Given a module category $\mathcal{M}$ over $\mathcal{C}$, the *dual category* $\mathcal{C}^*_{\mathcal{M}}$ is the category of functors that are $\mathcal{C}$-linear from $\mathcal{M}$ to $\mathcal{M}$, i.e $F(a \cdot i) \sim a \cdot F(i)$, where $\sim$ has to be intended as up to isomorphisms. With the tensorial product given by composition, under certain technical hypotheses, this category forms a fusion category. The easiest example encountered in the main text is given by the regular module category, where $\mathcal{M} = \mathcal{C}$, $\cdot = \otimes$. It is possible to prove that for this module category the dual category is $\mathcal{C}$ itself, with the physical interpretation of boundary as interfaces explained in the text.

B. Basics of 2D CFT

B.1 Virasoro algebra and its representations

Here we report some basic notions about 2D CFT useful for the main text. For an extended discussion see [14]. In a 2D Euclidean space, conformal symmetry is an extension of the isometry symmetry, which consists of all transformations that leave the metric invariant up to an overall function. In dimension $d > 2$, other than isometry, there are two new kinds of symmetries, the dilatation and the *special conformal transformations*, of which there are as many as there are translations. In 2D, because of the complex structure of the plane, it turns out that there are actually infinitely many transformations, whose vector fields follow the *Witt algebra* (B.1).

$$[L_n, L_m] = (n-m)L_{n+m}, \quad [\bar{L}_n, \bar{L}_m] = (n-m)\bar{L}_{n+m}, \quad [L_n, \bar{L}_m] = 0. \tag{B.1}$$

In the quantum formulation of the theory, $L_n, \bar{L}_m$ generate an operator algebra that, in general, satisfies the central extension of the Witt algebra, called *Virasoro algebra*. In particular we have two independent copies of it:

$$[L_n, L_m] = (n-m)L_{n+m} + \frac{c}{12}m(m^2-1)\delta_{n+m,0}, \quad [L_n, \bar{L}_m] = 0. \tag{B.2}$$

Here, c and $\bar{c}$ are called *central charges*. Next we are interested in representations of the Virasoro algebra. In order to do this, we consider the *primary states* $|h\rangle$ such that:

$$L_0 |h\rangle = h |h\rangle, \quad L_n |h\rangle = 0, \quad \forall n \geq 1. \tag{B.3}$$

The *descendants* are created by applying the L_{-n} operators to the primaries. This set of states generates the irreducible representations of the Virasoro algebra [3]. Notice that in conformal field theories there is a natural map between primary states in the Hilbert space of a circle and operators at the origin, called *primary operators*, which, roughly speaking, consist in associating with an operator the state obtained by shrinking the Hilbert space onto the origin by the $\exp(-\alpha D)$ operator and applying the operator at the origin to the vacuum. This relation is called *state-operator correspondence*.

B.2 Minimal models

One important step in classifying CFTs is to determine the content of primary operators. In most cases, these are infinite and the theory is difficult to manage. One family of exceptions to this is given by *minimal models* $\mathcal{M}_{p,q}$, where the number of primary fields is finite and given by the following central charge and conformal weights.

$$c = 1 - \frac{6(p-q)^2}{qp}, \quad h_{r,s} = \frac{(qr-ps)^2 - (p-q)^2}{4pq}, \quad p, q \in \mathbb{N}, \quad gcd(q,p) = 1. \tag{B.4}$$

[3] This is true only after quotienting by the *null states*, descendants $|\psi\rangle$ with the property that $L_n |\psi\rangle = 0$, $\forall n \geq 1$.

The number of primary fields is finite since $h_{r,s} = h_{r+p,s+q}$. The primaries satisfy the truncated OPE (neglecting dynamical factors and descendants):

$$\phi_{r,s} \times \phi_{m,n} = \sum_{\substack{|r-m|+1 \leq k \leq k_{\max} \\ k \equiv r+m+1 \pmod 2}} \sum_{\substack{|s-n|+1 \leq \ell \leq \ell_{\max} \\ \ell \equiv s+n+1 \pmod 2}} \phi_{k,\ell} \tag{B.5}$$

Here, $k_{\max} = \min(r+m-1,\, 2p-1-r-m)$, $l_{\max} = \min(s+n-1,\, 2q-1-s-n)$. We say that the minimal model is *unitary* if $p = q+1$, and *diagonal* if $c = \bar{c}$, $h_{r,s} = \bar{h}_{r,s}$. An example of diagonal minimal unitary model is given by the critical Ising model, $\mathcal{M}_{4,3}$, with three primaries:

$$\phi_{1,1} = \mathbb{I}, \quad \phi_{2,1} = \epsilon, \quad \phi_{2,2} = \sigma. \tag{B.6}$$

The conformal weights are:

$$c = \frac{1}{2}, \quad h_\epsilon = \frac{1}{2}, \quad h_\sigma = \frac{1}{16}. \tag{B.7}$$

These primaries follow the truncated OPE:

$$\sigma \times \sigma = \epsilon, \quad \sigma \times \epsilon = \sigma, \quad \epsilon \times \epsilon = 1. \tag{B.8}$$

These are the same as the fusion relations of the symmetry lines in the Ising category. In general, in diagonal minimal models $\mathcal{M}_{n+1,n}$ there is a bijection between symmetry fusion lines and primary operators. Considering these theories in terms of an action, we can deform the theory using a primary operator $\phi_{1,3}$, such that the deformed theory is integrable, and with a symmetry fusion category $\mathcal{A}_n$ described in (2.26).

C. S-matrix bootstrap and integrability

Here we clarify some general features of the S-matrix bootstrap in 2D for relativistic models, with and without integrability condition. For an extensive discussion see [9, 10, 15].

C.1 General features

Here we always consider scattering $2 \to 2$ of relativistic models and we try to construct the S-matrix for the lightest massive particles, with mass m. Since the models are relativistic, we use the Mandelstam variables s, t, u, which satisfy $s + t + u = 4m^2$. Since we assume that the momenta don't change in the scattering, we can set $u = 0$ and $t = 4m^2 - s$. The different conditions that our S-matrix should satisfy are:

- Analyticity. From causality arguments one can usually extend the S-matrix to the entire complex plane $S(s)$. The assumption we made is that it is analytic except for the lines $s \geq 4m^2$ and $s \leq 0$ from the scattering threshold of the fundamental particles, and for some pole singularities, corresponding to bound states.

- Unitarity. The S-matrix is a unitary operator that maps the IN and OUT states. Writing $\langle p_3, p_4 | p_1, p_2 \rangle = (2\pi)^2 \sqrt{s}\sqrt{s-4m^2}\delta^2(p_1 + p_2 - p_3 - p_4) S(s)$ we arrive at the constraint:

$$|S(s)|^2 \leq 1, \quad s \geq 4m^2. \tag{C.1}$$

  In the analytic extension, we can add the *real analyticity* condition $S^*(s) = S(s^*)$.

- Symmetries. A possible requirement that adds constraints in the bootstrap is that the S-matrix commutes with the generators of symmetries of our model, $[Q_s, S] = 0$. Q_s should be understood as the two particle representation of the generators of Lie algebra of the symmetry.

- Crossing equation. Once we have imposed analyticity, unitarity and symmetry constraints, there is a fourth property, related to the equivalence of the s-channel and t-channel, or also to the existence of a particular vacuum of the theory which commutes with all the particle representations. This is discussed in the text, but usually is written informally as $S(s) = S(4m^2 - s)$.

C.2 Integrability

The conditions described above are not sufficient to determine unequivocally the S-matrix of the theory. A technical, not generic and quite strong, assumption that does this job is the *Yang–Baxter equation*:

$$S_{12}(\theta_{12})\, S_{13}(\theta_{13})\, S_{23}(\theta_{23}) = S_{23}(\theta_{23})\, S_{13}(\theta_{13})\, S_{12}(\theta_{12}), \tag{C.2}$$

Here, $\theta_{ij} = \theta_i - \theta_j$ is the *rapidity* difference, related to s by $s = m_1^2 + m_2^2 + 2m_1 m_2 \cosh\theta_{ij}$. This equation can be derived by assuming that the momentum states are described by creation operators $A^\dagger$ that act on the vacuum Ω, and satisfy the Zamolodchikov–Faddeev algebra

$$A_a^\dagger(\theta_1)\, A_b^\dagger(\theta_2) = S_{ab}^{\ cd}(\theta_1 - \theta_2)\, A_d^\dagger(\theta_2)\, A_c^\dagger(\theta_1). \tag{C.3}$$

Physically, usually integrability comes from the presence of higher spin charged operators. Integrable models are much more constrained than non-integrable ones. For example, scattering processes of the type $2 \to n$, with $n \geq 3$, are not possible. This implies, by unitarity, that $|S(s)|^2 = 1$. Also, the mass of the particles must remain the same. There are different interesting models that are indeed integrable. As mentioned above one class of these models are the minimal models deformed by a primary operator.

C.3 Numerical bootstrap

If one does not consider integrability in the bootstrap, the best one can do is to obtain numerical bounds for fixed amplitudes. The unitarity bounds imply that the resulting region is convex. This allows one to use convex optimisation to determine the boundary of a region by extremising a linear functional. The specific procedure depends on the problem under consideration, in particular on how the S-matrix is decomposed, the presence of couplings corresponding to bound states, and the type of functional adopted. A particular choice of this functional is made in the main text.

D. SymTFT formalism and modified crossing refined

In the SymTFT framework we consider our 1+1 theory as a boundary of a 2+1-dimensional bulk, to which we add another boundary on the opposite side. This is encoded in the triplet:

$$(\mathcal{Z}(\mathcal{C}), \mathcal{L}, X_{rel}). \tag{D.1}$$

For details of each element, we refer to [13]. A boundary in the two-dimensional theory extends to a boundary in the bulk. Whereas symmetry lines are confined to the original theory, punctures and dual lines extend into the bulk, as shown in Figure 31.

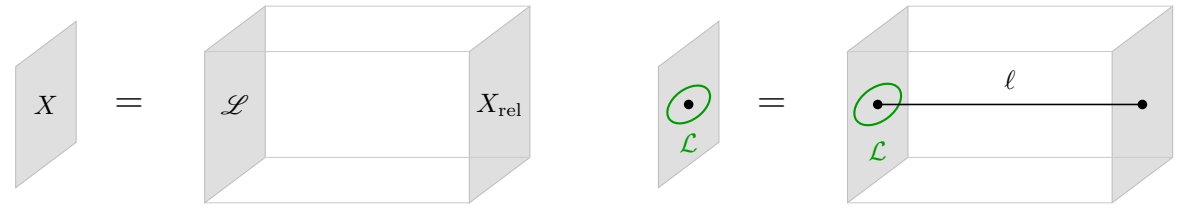


Figure 31: TFT representation of 2d spacetime and a symmetry line.

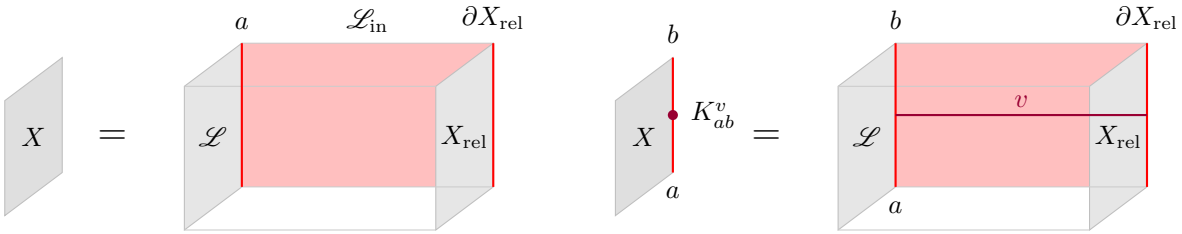


Figure 32: TFT representation of module category.

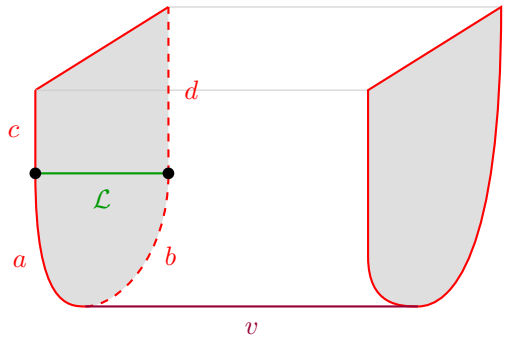


Figure 33: TFT representation of the isomorphism shown in Figure 19.

In order to introduce the S-matrix in this framework, we can represent it in the same way discussed in the previous section at the free boundary, as in Figure 34.

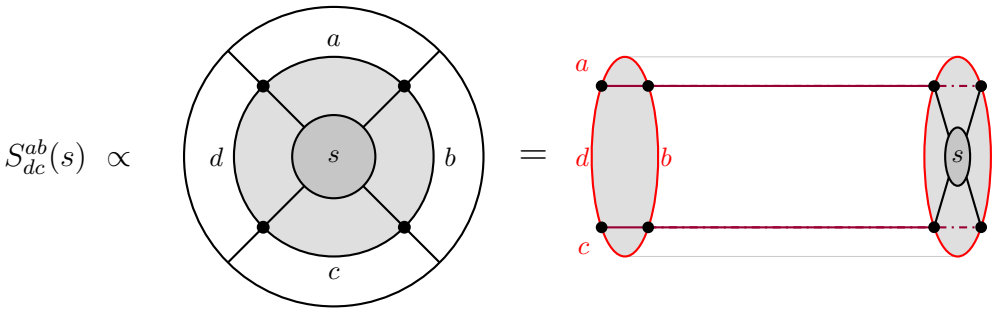


Figure 34: TFT representation of S-matrix.

In the same way we can implement the IN and OUT states normalisation (Figure 36).

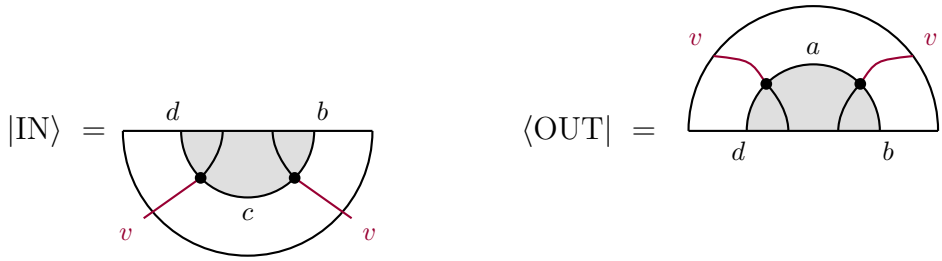


Figure 35: Diagrammatic 2D representation of IN and OUT states.

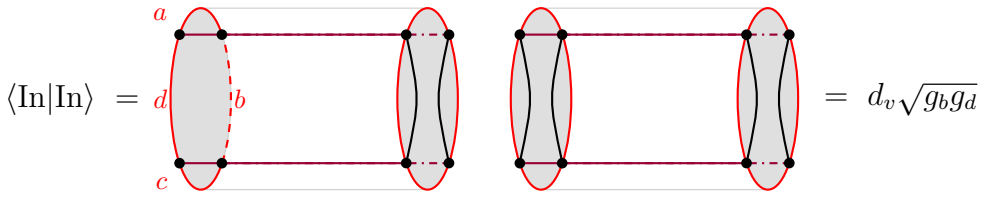


Figure 36: TFT representation of IN normalisation.

The calculation is straightforward, as shown in Figure 37, and gives the generalisation of the crossing relation for a general module category.

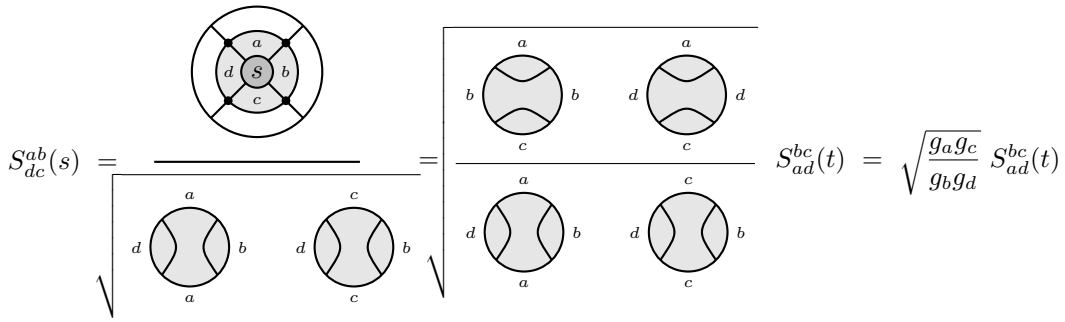


Figure 37: Diagrammatic proof of modified crossing relation in any representation.

In the same spirit, by inserting the S-matrix on the second boundary, we can derive the Ward identities in the original theory by deforming a symmetry line in two distinct ways (see Figure 38).

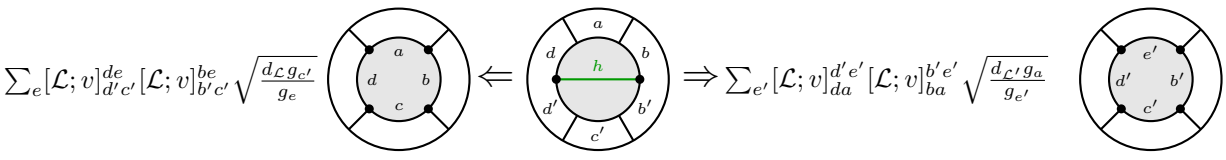


Figure 38: Diagrammatic proof of Ward identity.

Matching results in Figure 38, we obtain the result:

$$\sum_e [\mathcal{L};v]^{de}_{d'c'}[\mathcal{L};v]^{be}_{b'c'}\sqrt{\frac{g_c g_b g_d}{g_e}}\, S^{ab}_{de}(s) = \sum_{e'} [\mathcal{L};v]^{d'e'}_{da}[\mathcal{L};v]^{b'e'}_{ba}\sqrt{\frac{g_a g_{b'} g_{d'}}{g_{e'}}}\, S^{e'b'}_{d'c'}(s). \tag{D.2}$$